Ch.1 Basic concepts:

section (1) : Introduction :

An electric circuit : is an interconnection of electric elements.

section (2) : system of units:

Team
Zerone
Computer & Network Engineers

Electric current ampere A
charge coulomb C

The SI prefixes

10^9	giga	G	10^6	mega	M
10^3	kilo	K	10^2	hecto	h
10^{-6}	Micro	μ	10^{-9}	nano	n
10^{-12}	pico	p			

Section (3) : charge and current :

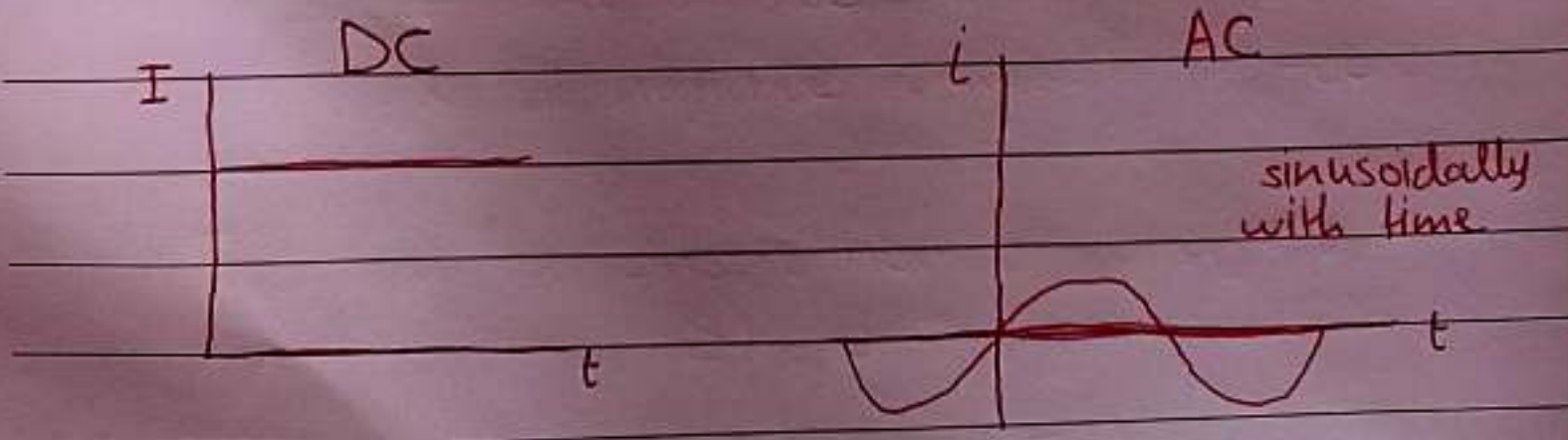
$1C \longrightarrow 6.24 \times 10^{18}$ electron save

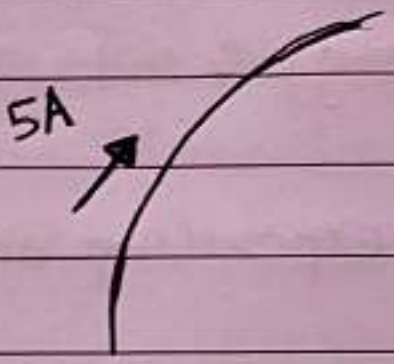
* The law of conservation of charge :
states that charge can neither be created nor destroyed, only transferred. Thus the algebric sum of the electric charges in a system doesn't change.

In the conductor the positive and negative charges are move.

$$i = \frac{\Delta q}{\Delta t} \qquad A = \frac{C}{S} \qquad Q = \int_{t_0}^{t} i \, dt$$

$q = I \Delta t$





positive current flow

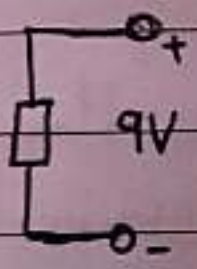


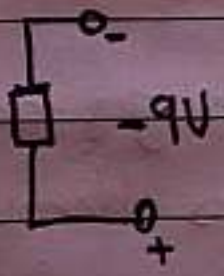
negative current flow

section(4) : voltage :

Team ZerOne Computer & Network

$$V_{ab} = \frac{dw}{dq} \quad \leftarrow \text{energy J}, \quad \leftarrow \text{c}$$





* دائماً بيتحكم بالإشارة q عند كمية الشحنة تكون إشارة.

section(5) : power and energy :

Power : is the time rate of expending or absorbing energy.

$1\,Wh \rightarrow 3600\,J$

$$p = \frac{dw}{dt} = vi$$

* The power called the (instantaneous power
* The power absorbed or supplied

Team ZeroOne Computer & Networks Engineers

* $p = +ve \rightarrow$ absorbed
  $p = -ve \rightarrow$ supplied

* passive sign convention : when the current enters through the positive terminal of an element and $p = +ve$. If the current enters through the negative terminal $p = -ve$.

The law of conservation of energy :

$$\Sigma P = 0$$

$$P_{supply} + P_{absorbed} = 0$$

* The energy absorbed or supplied by an element

$$w = \int_{t_0}^{t} p\,dt = \int_{t_0}^{t} vi\,dt$$

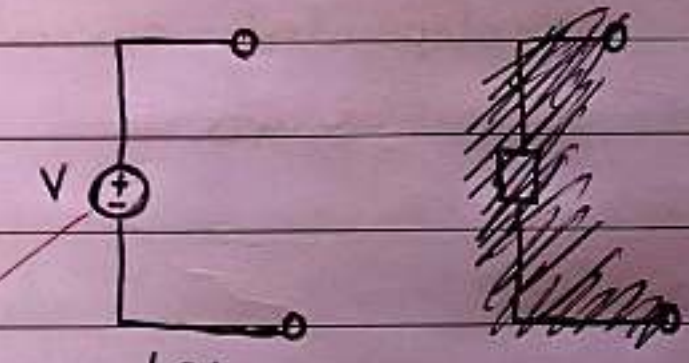
* Energy is the capacity to do work

J = [illegible] watt-hour

chapter(6) : circuit elements :

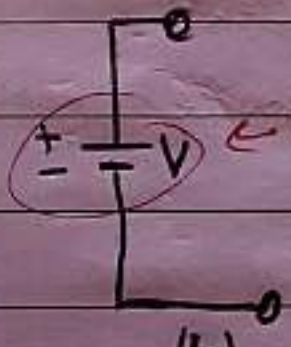
• An ideal independent source is an active element that provides a specified voltage or current that is completely independent of other circuit elements.

Zerone team
Computer & Networks Engineers



used for constant or time-varying voltage.

→ maybe be AC or DC and this symbol denote to the



used for constant voltage (dc)

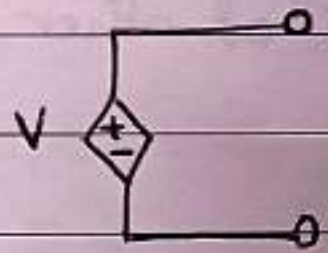
DC [illegible]

AC [illegible]

- An ideal dependent (or controlled) source is an active element in which the source quantity is controlled by another voltage or current.

- types of dependent sources, namely:
  1. A voltage-controlled voltage source (VCVS)
  2. A current- " " " (CCVS)
  3. A voltage-controlled current source (VCCS)
  4. A current- " " " (CCCC)

Team ZeroOne Computer & Networks Engineers



dependent voltage source



dependent current source

Ch.2 Basic laws:

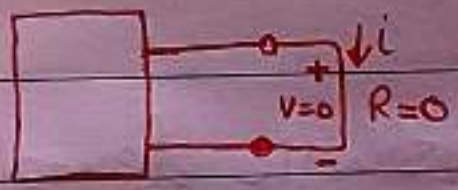
Section(1): Ohm's law:

resistance $R = \frac{\rho L}{A}$

resistivity ρ ← constant

$$v = iR$$

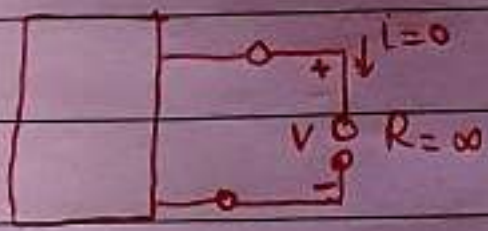
- A short circuit: is a circuit element with resistance approaching zero.



$R = 0 \quad v = iR = 0$

i (maximum)

- An Open circuit: is a circuit element with resistance approaching infinity.



$R = \infty \quad i = 0 = V_L = 0$

v (maximum)

power dissipated by a resistor

$$p = vi = i^2R = \frac{v^2}{R}$$

- The power non linear function of either current or voltage, ohm's law linear $v = iR$



- R is positive quantites, the power dissipated in a resistor is always +ve, a resistor always absorbs power from the circuit. This confirms the idea that a resistor is a passive element, incapable of generating energy.

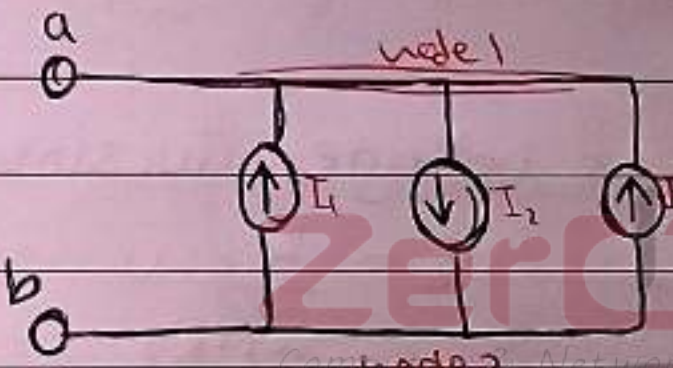
section(2) : Nodes, Branches, loops :

- branch represents a single element
- Node is the point connection between two or more branches.
- Loop is any closed path in a circuit.

section(3) : kirchhoff's laws : KCL

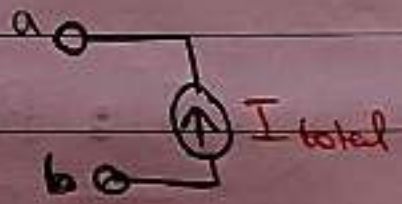
: states that the algebraic sum of the currents entering a node is equal to the sum of the currents leaving the node

$$\Sigma i_{entering} = \Sigma i_{leaving}$$



by KCL

$$I_T = I_1 - I_2 + I_3$$
$$= I$$



: kirchhoff's voltage law : KVL

: states that the algebraic sum of all voltages around a closed path (or loop) is zero.

section(4): series Resistor and voltage Division:

The equivalent resistance of any number of resistor connected in series is the sum of individual resistances.

$$R_{eq} = R_1 + R_2 + \cdots + R_n = \sum_{n=1}^{N} R_n$$

principle of voltage division:

$$v_1 = \frac{R_1}{R_1+R_2} v \qquad v_2 = \frac{R_2}{R_1+R_2} v$$

Zeroone Computer & Networks Engineers

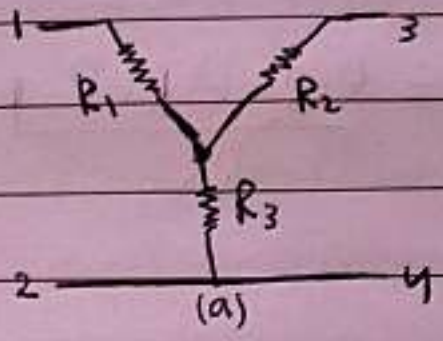
section(5): Parallel Resistors and current Devision:

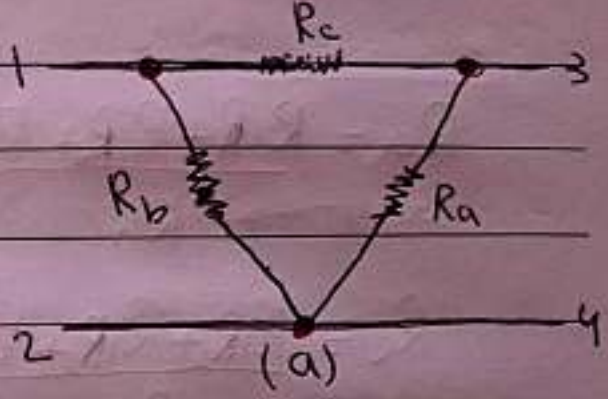
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

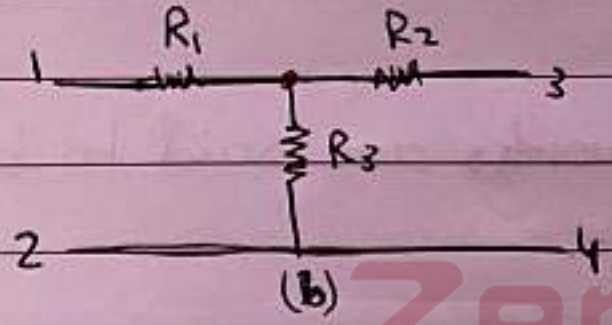
$$i = i_1 + i_2 = \frac{v}{R_{eq}}$$

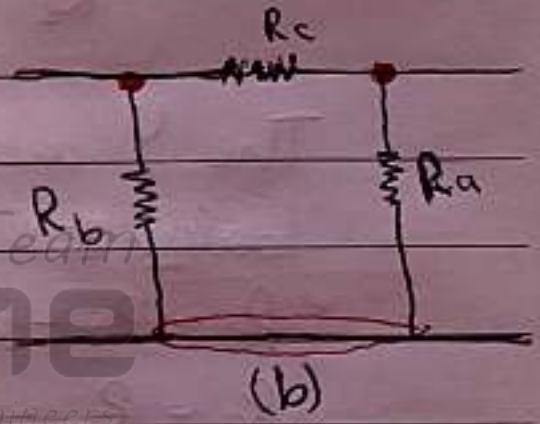
$$i_1 = \frac{R_2 i}{R_1+R_2} \qquad i_2 = \frac{R_1 i}{R_1+R_2}$$

section (6): Wye-Delta Transformations:





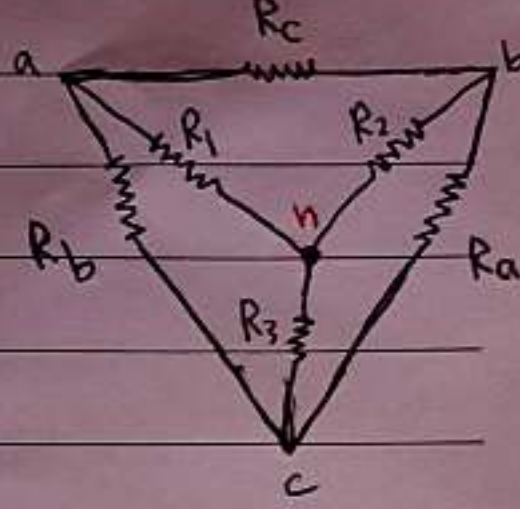




Two forms of the same network a) Y b) T

Two forms of the same network a) Δ b) Π

* Delta to Wye Conversion:
(Ra, Rb, Rc) given



$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} \qquad R_2 = \frac{R_c R_a}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

* Wye to Delta conversion:

(R_1, R_2, R_3) given

$$R_a = \frac{R_1R_2 + R_1R_3 + R_2R_3}{R_1} \qquad R_b = \frac{R_1R_2 + R_1R_3 + R_2R_3}{R_2}$$

الأبعد $\rightarrow R_1$

$$R_c = \frac{R_1R_2 + R_1R_3 + R_2R_3}{R_3}$$

The Y and Δ networks are said to be balanced when

$$R_1 = R_2 = R_3 = R_Y$$

$$R_a = R_b = R_c = R_\Delta$$

$$\boxed{R_Y = \frac{R_\Delta}{3}} \rightarrow \boxed{R_\Delta = 3R_Y}$$

Team Zeroone Computer Networks Engineers

Ch. 3 Methods of Analysis :

[Nodal Analysis]

- Steps to Determine node voltages :

1. select a node as the reference node
2. Apply KCL to each of the n-1 nonreference nodes . Use ohm's law to express the branch currents in terms of node voltages .
3. solve the resulting simultaneous equations to obtain the unknown node voltages.

Team Zero ne Computer & Networks Engineers

• symbols of a reference node :



[Nodal analysis with voltage source]

• **Case 1**: if a voltage source is connected between the reference node and a non-reference node, we simply set the voltage at the non-reference node equal to the voltage of the voltage source.

v_1 = voltage

• **Case 2**: if the voltage source (dependent or independent) is connected between two non-reference node, the two non reference nodes form a **super node**. A supernode is formed by enclosing a (dependent or independent) voltage source connected between two non-reference nodes and any elements connected in parallel with it.

- remove the voltage source.
- write equation of node 2 and node 3 in same equation.
- the make equation from the super node

Team Zeroma Computer & Networks Engineering

[Mesh Analysis]

A mesh is a loop with doesn't contain any other loops within it

- Use mesh currents as the circuit variables

- steps to Determine Mesh currents:

1. Assign mesh currents $i_1, i_2, \ldots, i_n$ to the n Meshes.
2. Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
3. Solve the resulting n simultaneous equations to get the mesh currents.

[Mesh analysis with current sources]

• **Case 1**: when a current source exists only in one mesh, we set $i_2 = -5\,A$ and write a mesh equation for the other mesh in the usual way.



ZeroOne
Computer & Networks Engineers

• **Case 2**: when a current source exists between two meshes. we create a supermesh by excluding the current source and any elements connected in series with it.

A supermesh results when two meshes have a (dependent or independent) current source in common.

[nodal and mesh analysis by inspection]

- Mesh analysis voltage source.
- Nodal analysis current source.

Nodal analys

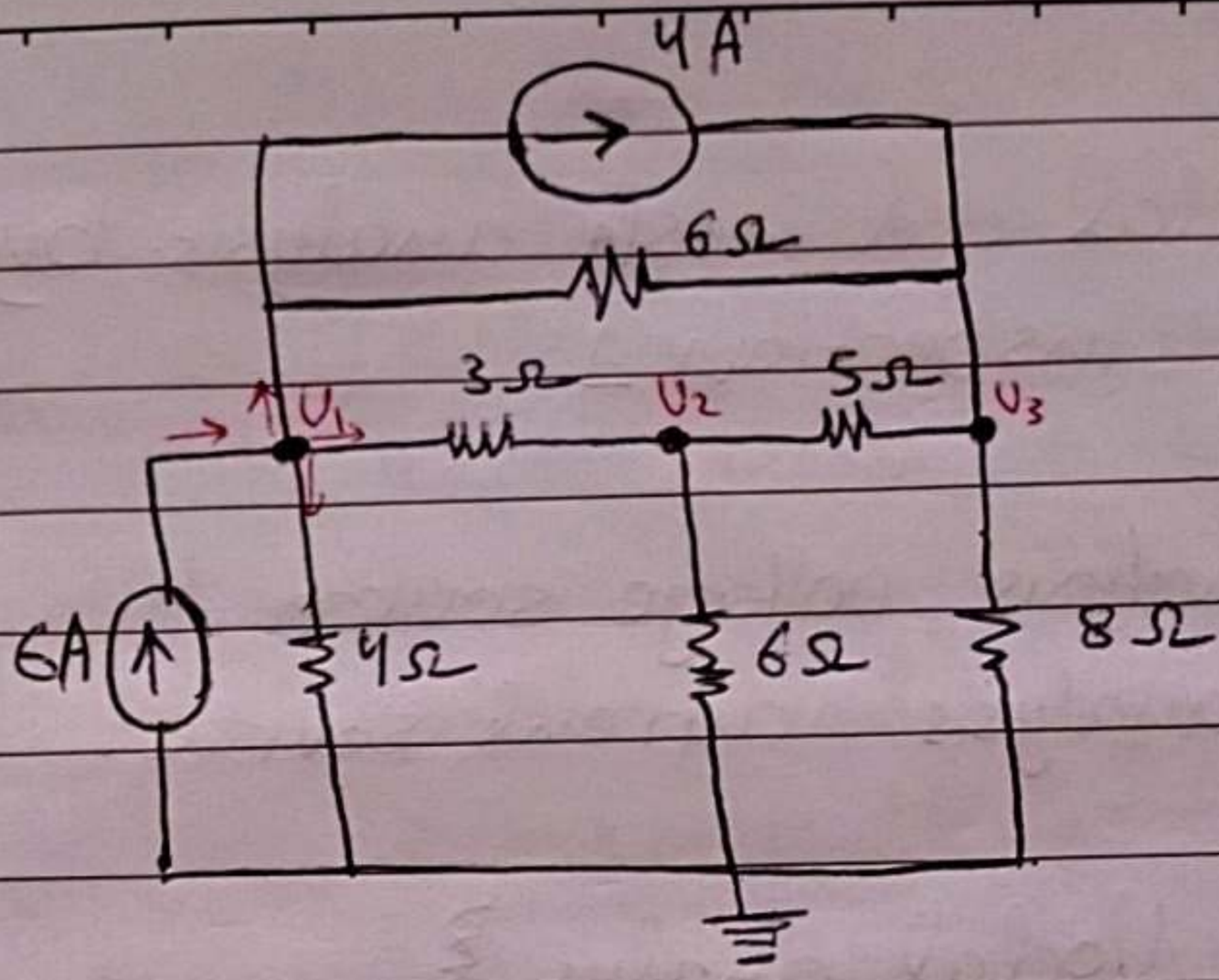
1) Find a reference node with known voltage
2) Assign voltages to other nodes
3) Assign conductance to each resistor
4) Find the elements of the conductance matrix
   a) for each node Gii
   b) between each 2 nodes $G_{ij} = G_{ji}$
5) set up the node equations in matrix form
6) solve the node-voltage equations.

$$G = \frac{1}{R}$$

Nodal analysis

$I = V G$

$G = \frac{1}{R}$



$$\begin{bmatrix} \frac{3}{4} & -\frac{1}{3} & -\frac{1}{6} \\ -\frac{1}{3} & 0.7 & -\frac{1}{5} \\ -\frac{1}{6} & -\frac{1}{5} & \frac{59}{120} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 4 \end{bmatrix}$$

det بگیر

1) $\begin{bmatrix} 2 & -\frac{1}{3} & -\frac{1}{6} \\ 0 & 0.7 & -\frac{1}{5} \\ 4 & -\frac{1}{5} & \frac{59}{120} \end{bmatrix}$ ← det بگیر

2) $\begin{bmatrix} \frac{3}{4} & 2 & -\frac{1}{6} \\ -\frac{1}{3} & 0 & -\frac{1}{5} \\ -\frac{1}{6} & 4 & \frac{59}{120} \end{bmatrix}$ ← det بگیر

3) $\begin{bmatrix} \frac{3}{4} & -\frac{1}{3} & 2 \\ -\frac{1}{3} & 0.7 & 0 \\ -\frac{1}{6} & -\frac{1}{5} & 4 \end{bmatrix}$ ← det بگیر

$$V_1 = \frac{①}{D}$$

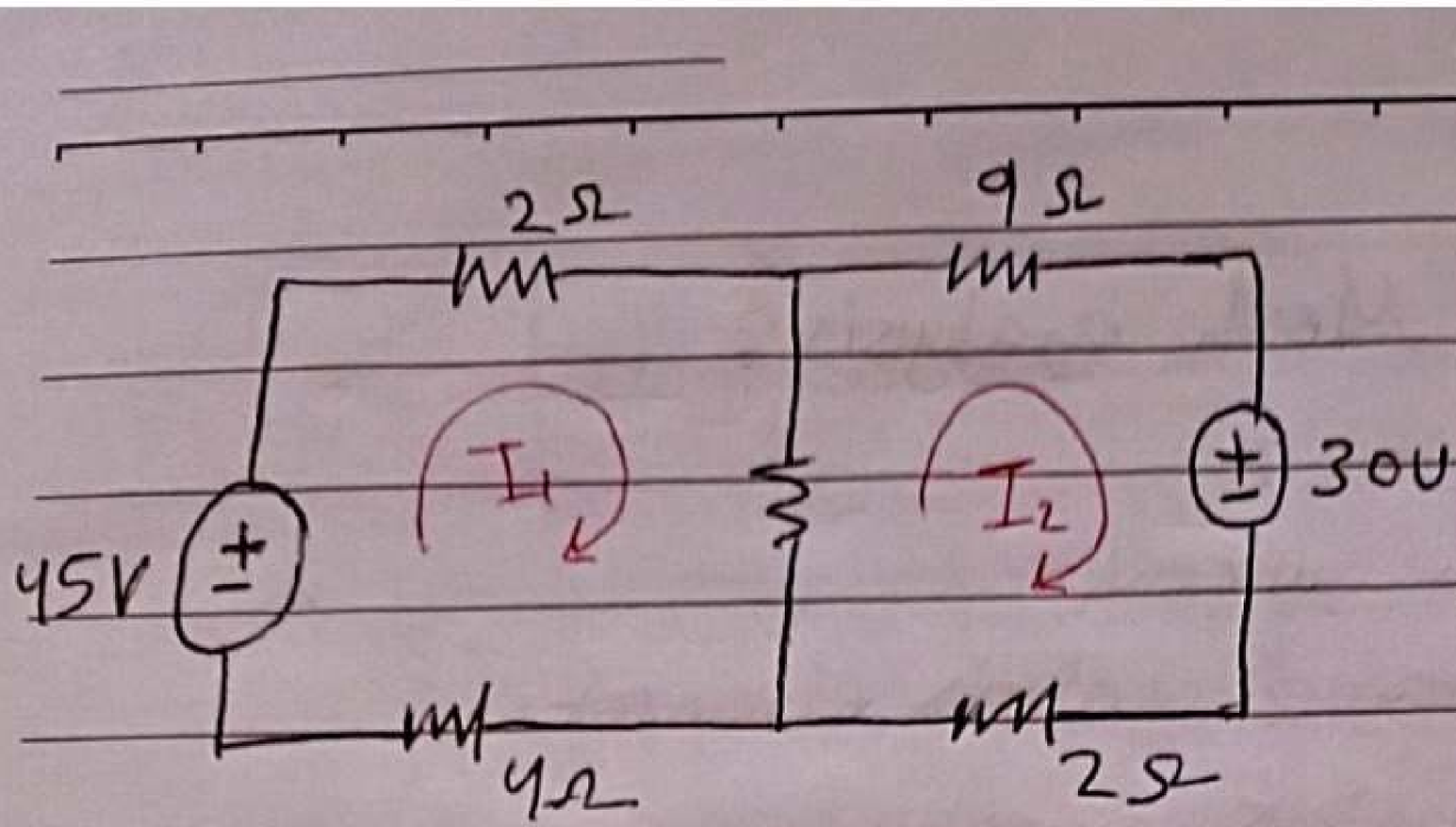
$$V_2 = \frac{②}{D}$$

$$V_3 = \frac{③}{D}$$

Mesh analysis

1) assign mesh currents
2) find resistance matrix elements
   a) $R_{ii} = \sum R$ mesh
   b) $R_{ij} = -R$ shared by i adj
3) find mesh voltage (in the direction of the mesh current, sources only)
4) set up the equations in matrix form and solve

Team
ZerOne
Computer & Network Company



mesh analysis

$U = IR$

$$\begin{bmatrix} 18 & -12 \\ -12 & 23 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 45 \\ -30 \end{bmatrix}$$

let this ↙

$= 270$

1) $\begin{bmatrix} 45 & -12 \\ -30 & 23 \end{bmatrix} \Rightarrow det = 675$

2) $\begin{bmatrix} 18 & 45 \\ -12 & -30 \end{bmatrix} \Rightarrow det = 0$

$$I_1 = \frac{①}{D} = \frac{675}{270} = 2.5A$$

$$I_2 = \frac{②}{D} = \frac{0}{270} = 0A$$

Ch.4 Circuit Theorems :

Section(1) : superposition :

- linear circuit: is one whose output is linearly related to its input.

- Superposition : principle states that the voltage across(or current through) an element in a linear circuit is the algebraic sum of the voltage across (or current through) that element due to each independent source acting alone.

- The principle of superposition helps us to analyze a linear circuit with more than one independent source.

- to apply superposition principle, we must keep to things in mind:

1) we consider one independent source at a time while all other independent sources are turned off. This implies that we replace every voltage source by a short circuit, and every current source by an open circuit.

2) Dependent sources are left intact because they are controlled by circuit variables.

- Steps to Apply superposition principle:

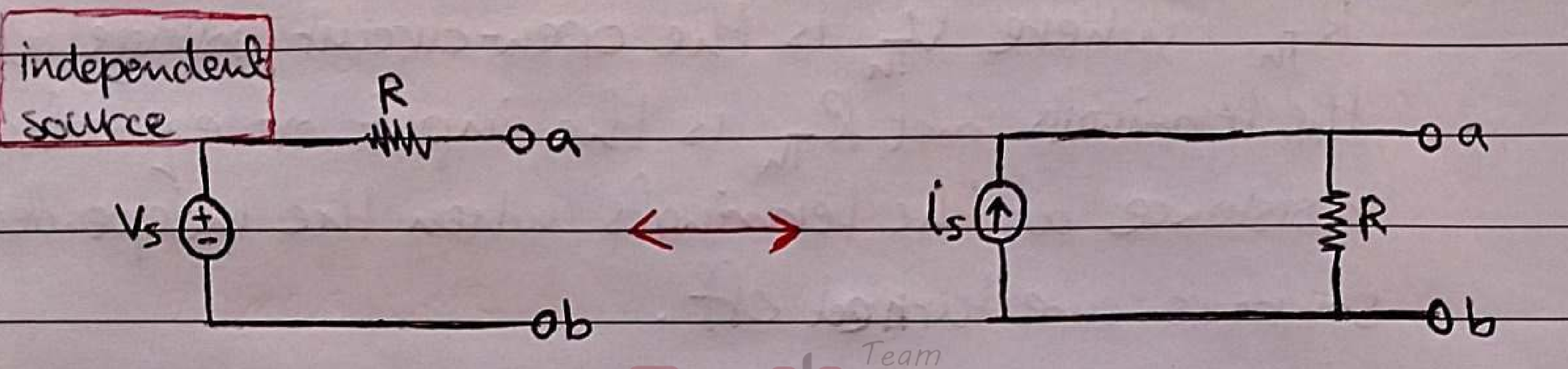
1) Turn off all independent sources expect one source. Find the output (voltage or current) due to that active source using the techniques covered in ch.2 and ch.3

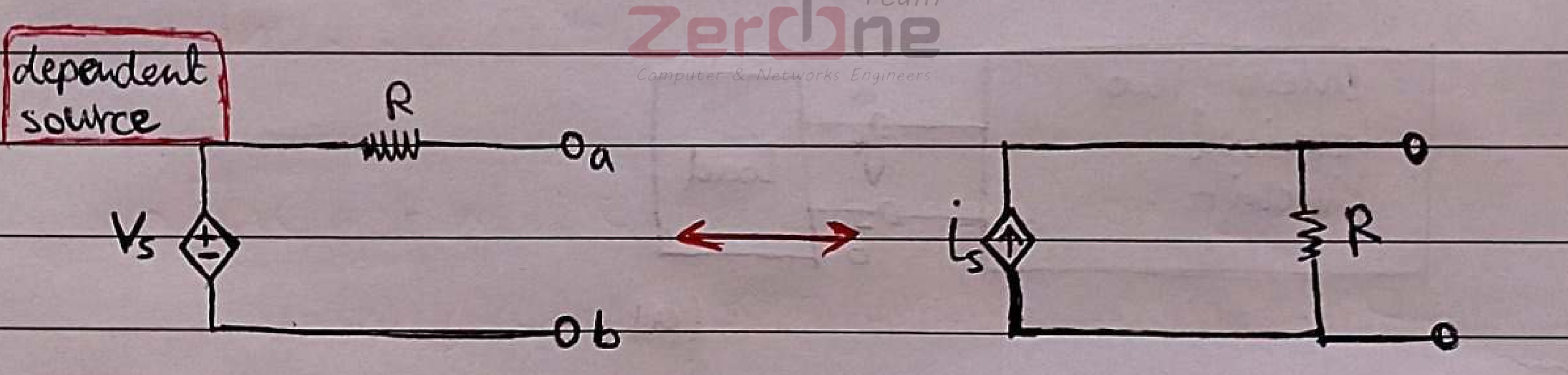
2) Repeat step1 for each of the other independent sources.

3) Find the total contribution by adding algebraically all the contributions due to the independent sources.

Section (2): source transformation:

- source transformation: is the process of replacing a voltage source V_s in series with a resistor R by a current source i_s in parallel with a resistor R, or vice versa

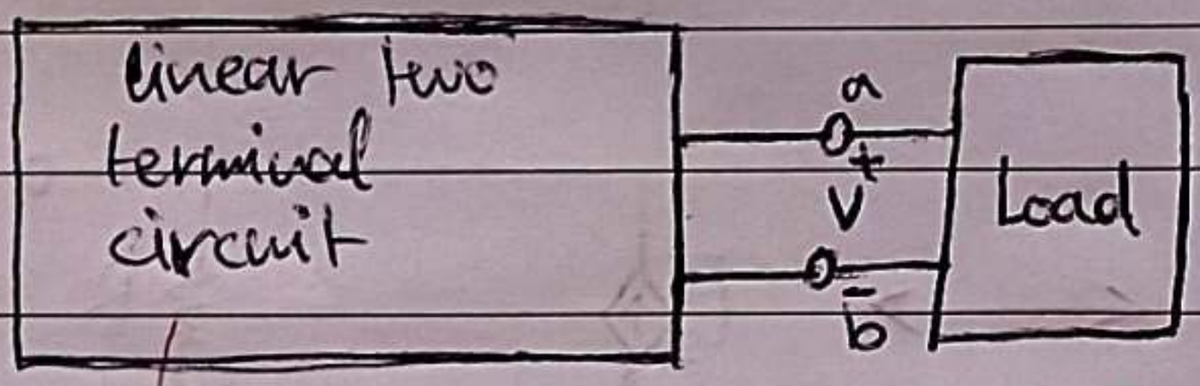




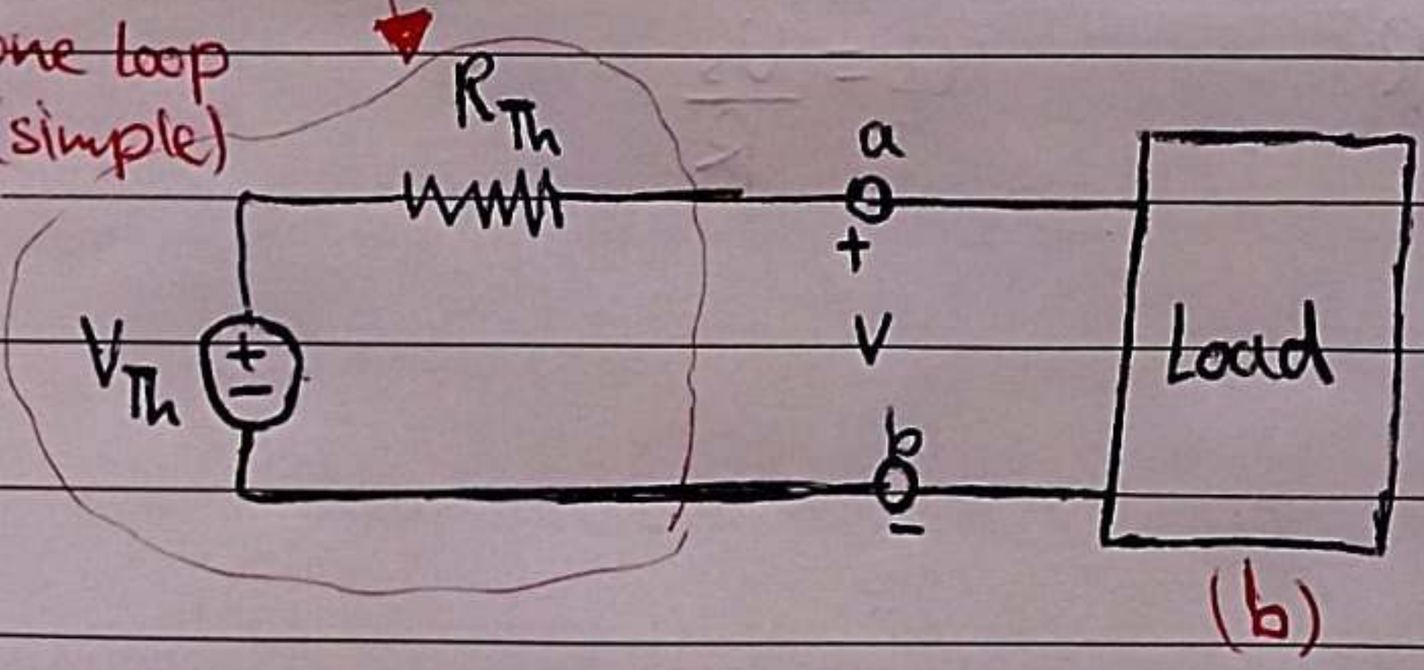
$V_s = i_s R$ $\qquad$ $i_s = \frac{V_s}{R}$

Section(3): Thevenin's Theorem:

- Thevenin's Theorem: states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th}, where V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.



(a)

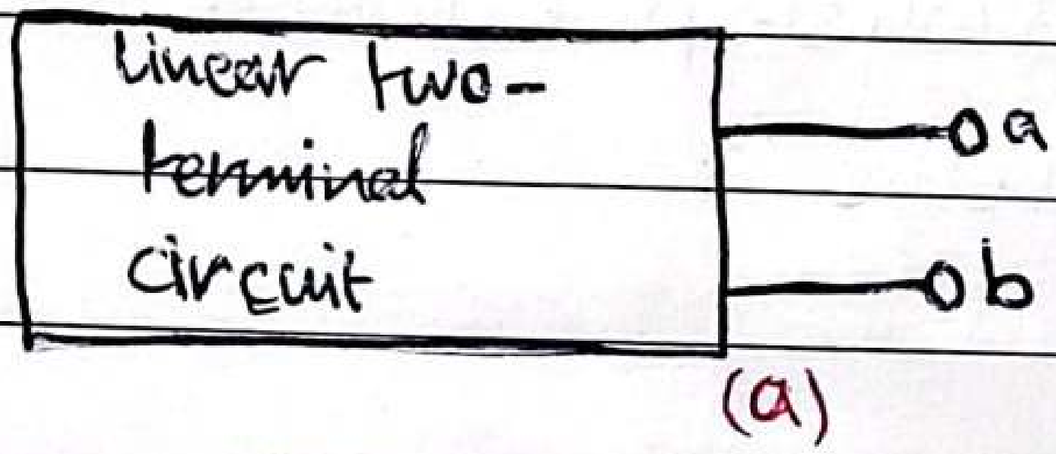


(b)

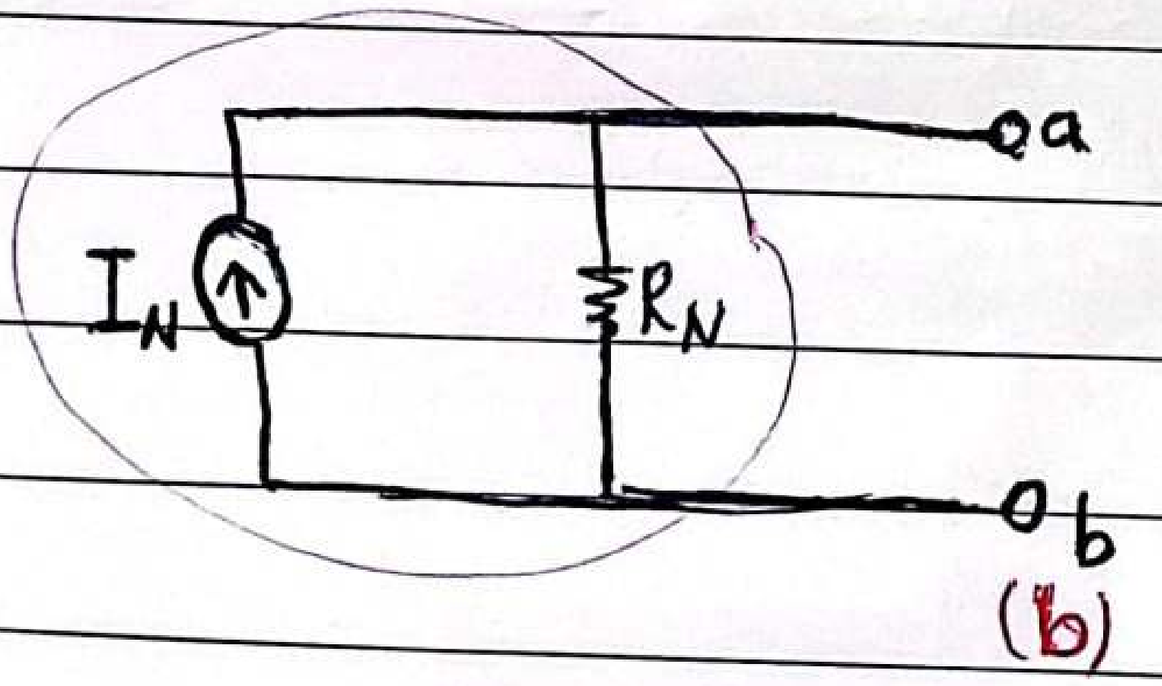
section (4): Nortons Theorem:

Nortons Theorem: states that a linear two-terminal circuit can be replaced by the equivalent circuit consisting of a current source I_N in parallel with a resistor R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.

Team ZerOne
Computer & Networks Engineers



(a)



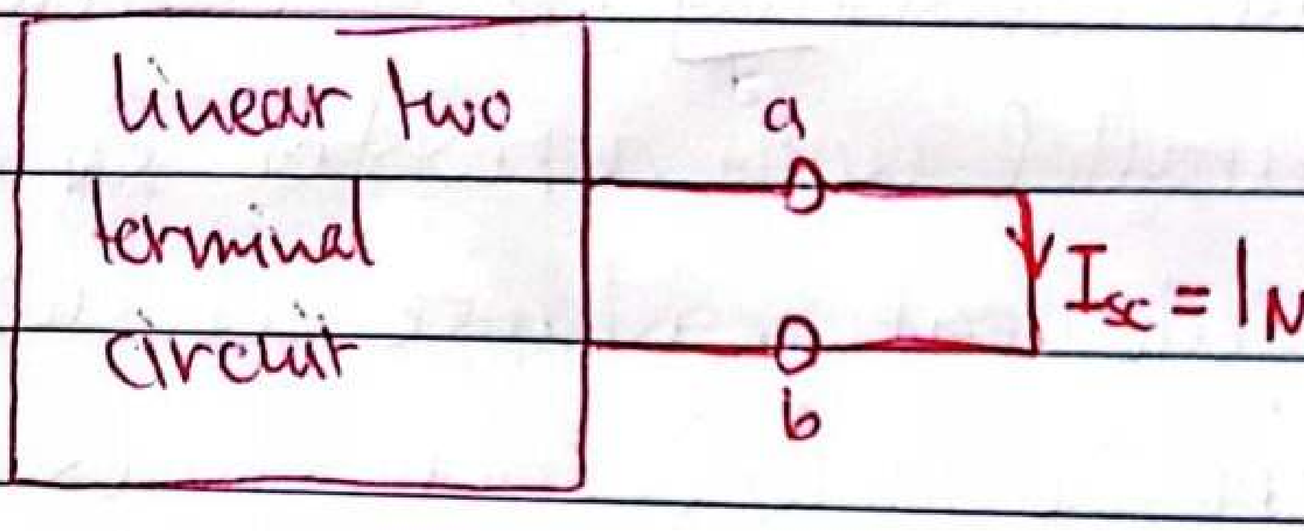
(b)

The Thevenin and Norton resistance are equal; that is

$$R_N = R_{Th}$$

• To find the Norton's current I_N :

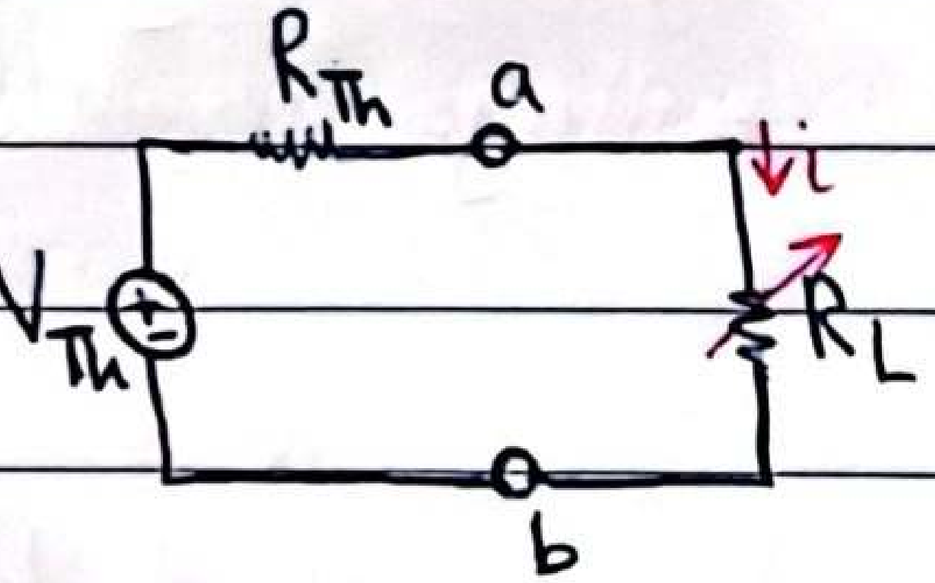
1 determine the short circuit current flowing from terminal a to b in both circuits



2 Observe the close relationship between Norton's and Thevenin's theorems

$$I_N = \frac{V_{TH}}{R_{TH}}$$

section(5): Maximum transfer power:

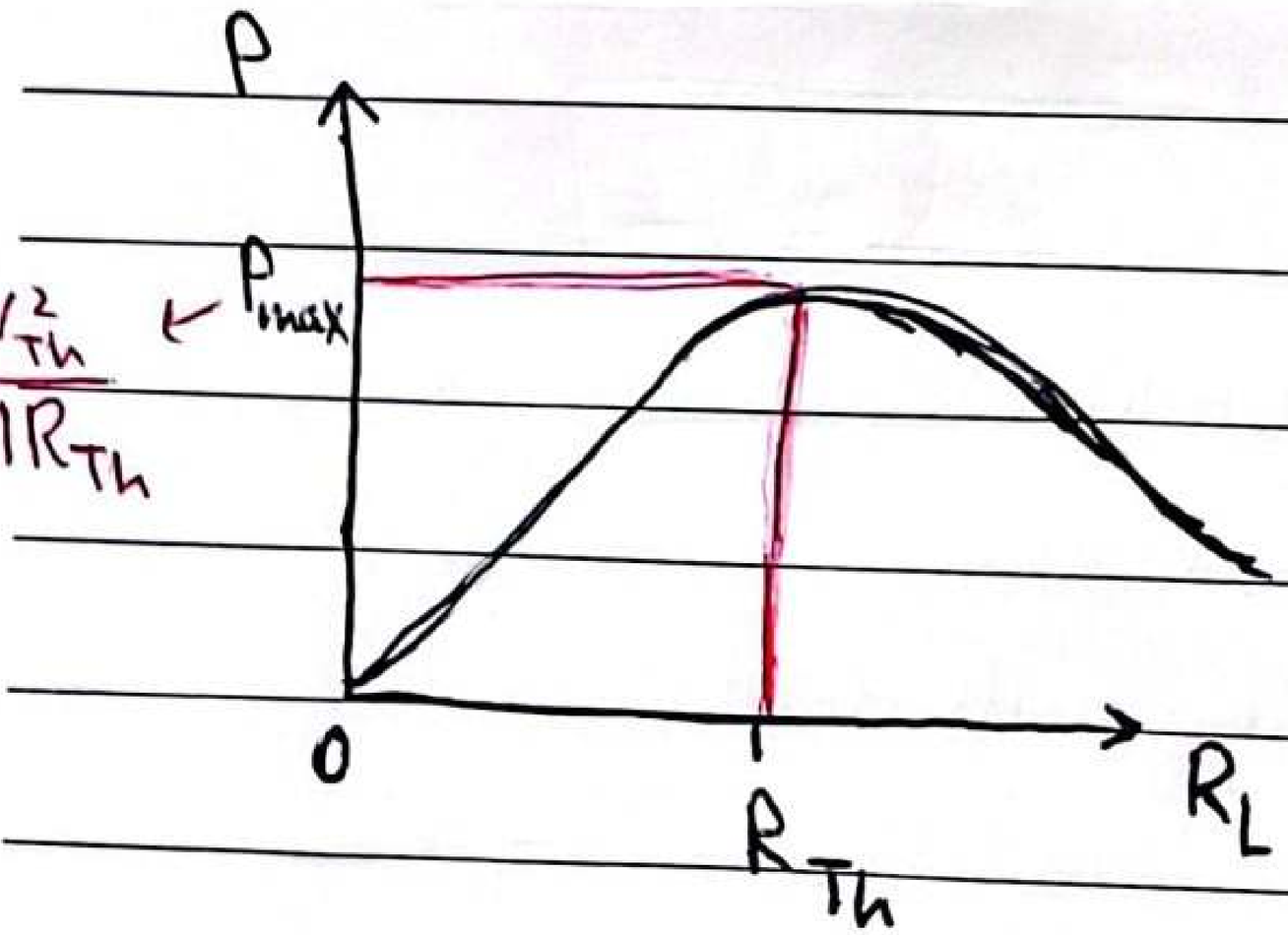


Maximum power: is transferred to the load when the load resistance equals the Thevenin resistance as seen from the load ($R_L = R_{Th}$)

$$i = \frac{V_{Th}}{R_{Th} + R_L}$$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

Team ZeroOne Computer & Networks Engineers



Ch.6 Capacitors and Inductors:

Capacitors:

1) is a passive element designed to store energy in its electric field.
2) used extensively in electronics, communic computers, and power systems.

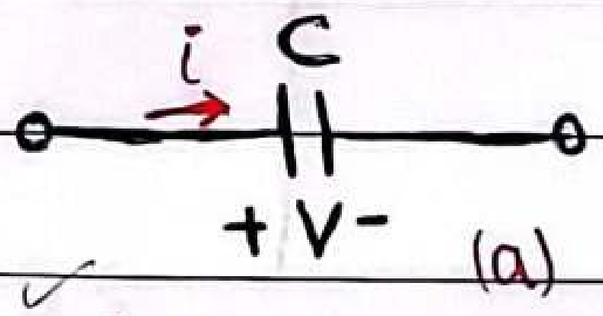
Team ZerOne Computer & Networks Engineers

A capacitor: consists of two conducting plates seperated by an insulator or dielectric.

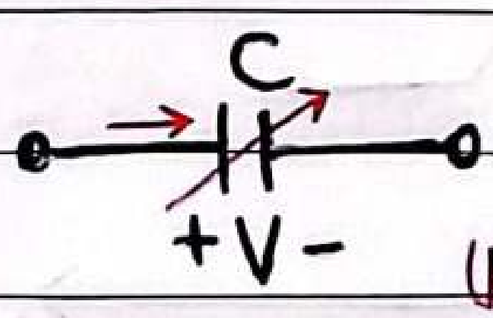
$$C = \frac{\epsilon A}{d}$$

$$q = CV \rightsquigarrow C = \frac{q}{V}$$

Farad



fixed capacitor



variable capacitor

$$i = C\frac{dv}{dt}$$

$$v(t) = \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)$$

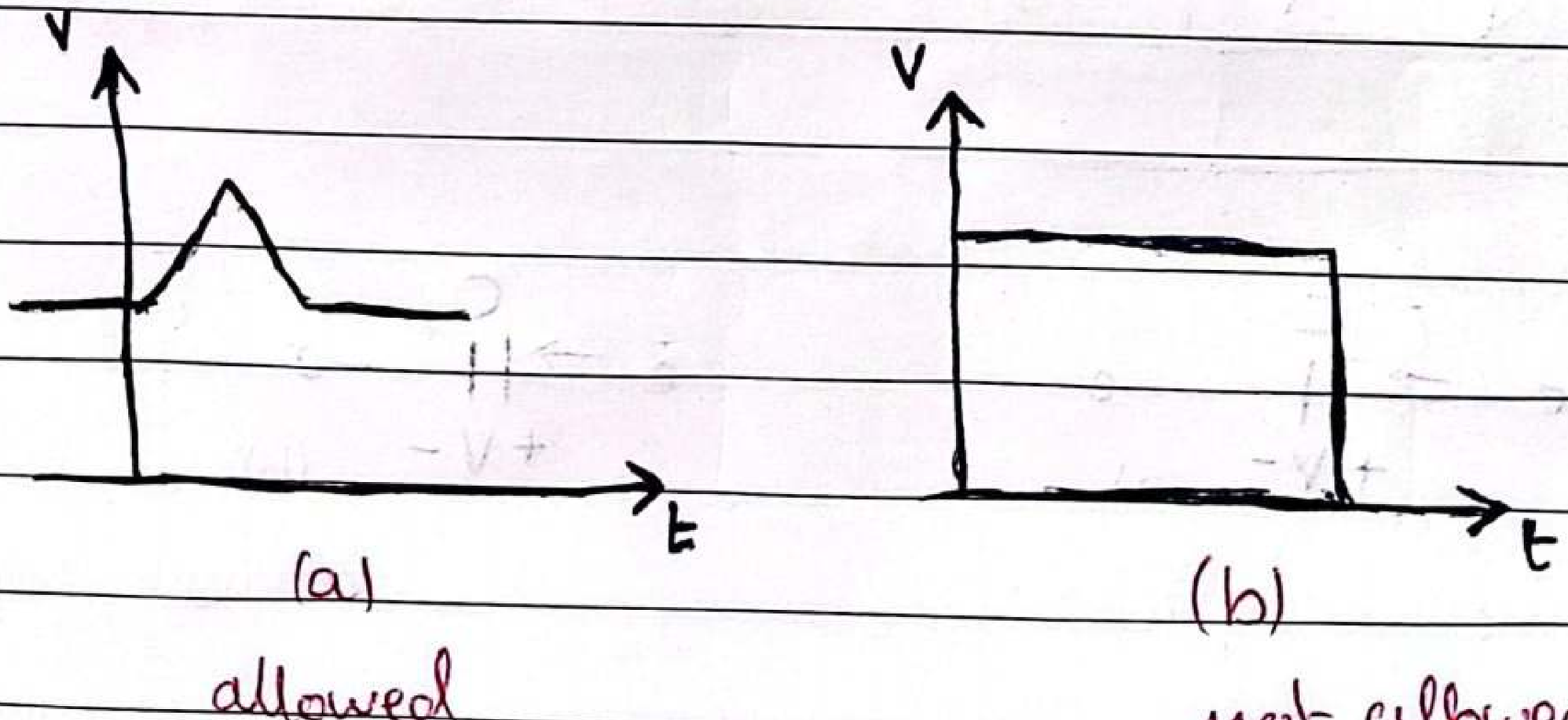
$i(t) = at + b$

$w = \frac{1}{2}Cv^2$ ← the energy stored in the electric field.

ZerOne TECHNO Computer & Networks Engineers

note:

- A capacitor is an open circuit to dc.
- The voltage on a capacitor cannot change abruptly.



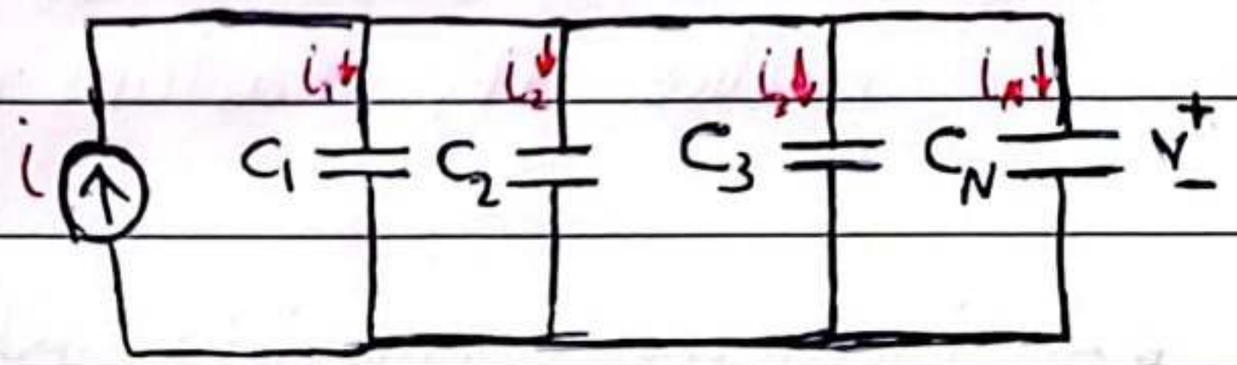
(a) allowed

(b) not allowed

- initially uncharged → $v(t)=0$
- initial capacitor voltage is zero → $v(t)=0$

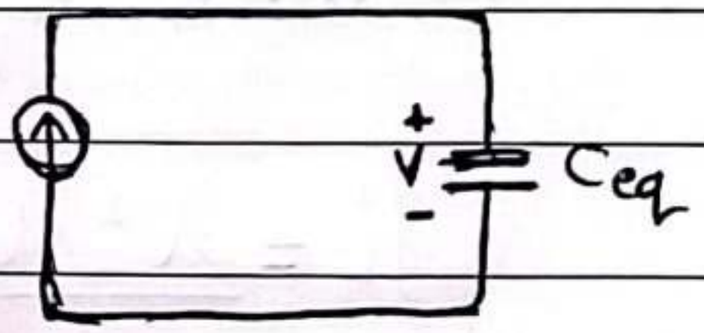
series and parallel capacitors:

[1] Parallel capacitors



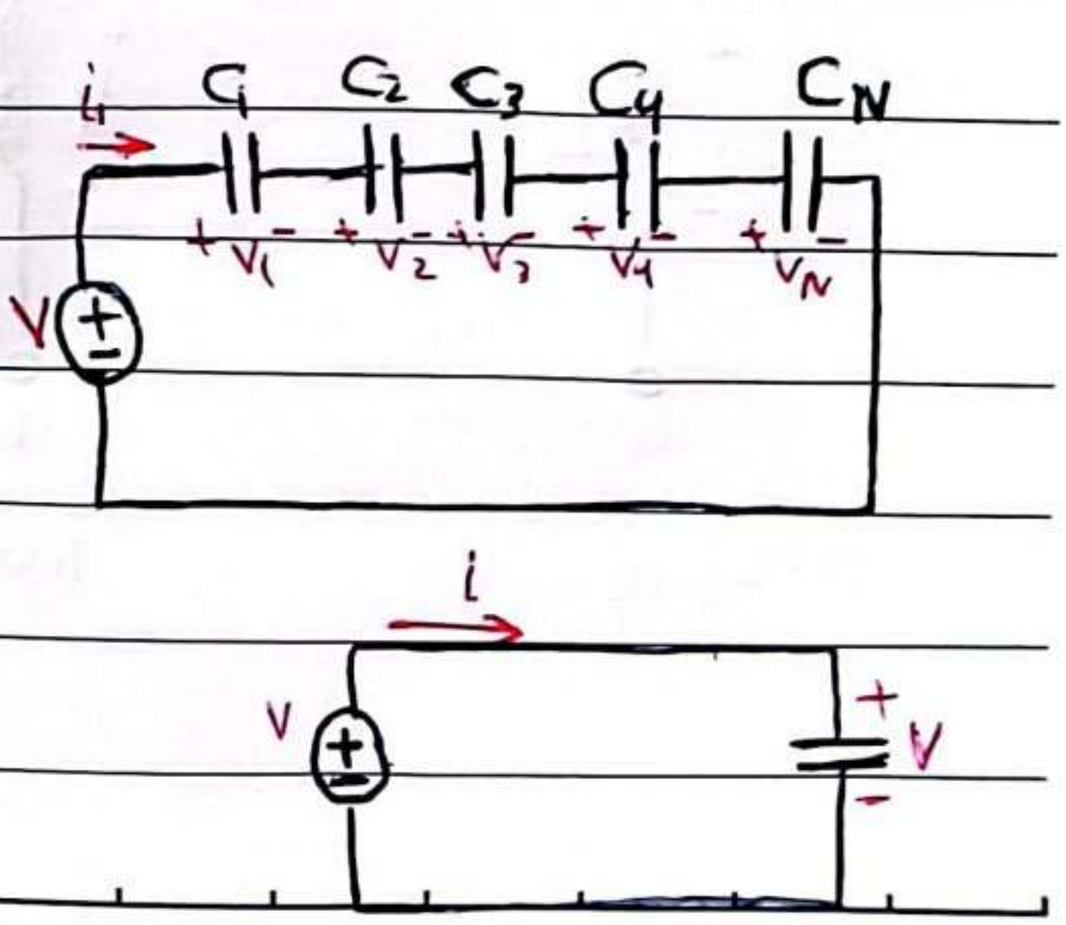
Team
ZerOne
Computer & Networks Engineers

$$C_{eq} = C_1 + C_2 + C_3 + \cdots + C_N$$



[2] series capacitors

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_N}$$



* اي تحزين ، الموت بيروح اي تحزين series مع الموت بيفضل parallel مع الموت بيفضل

Inductors :

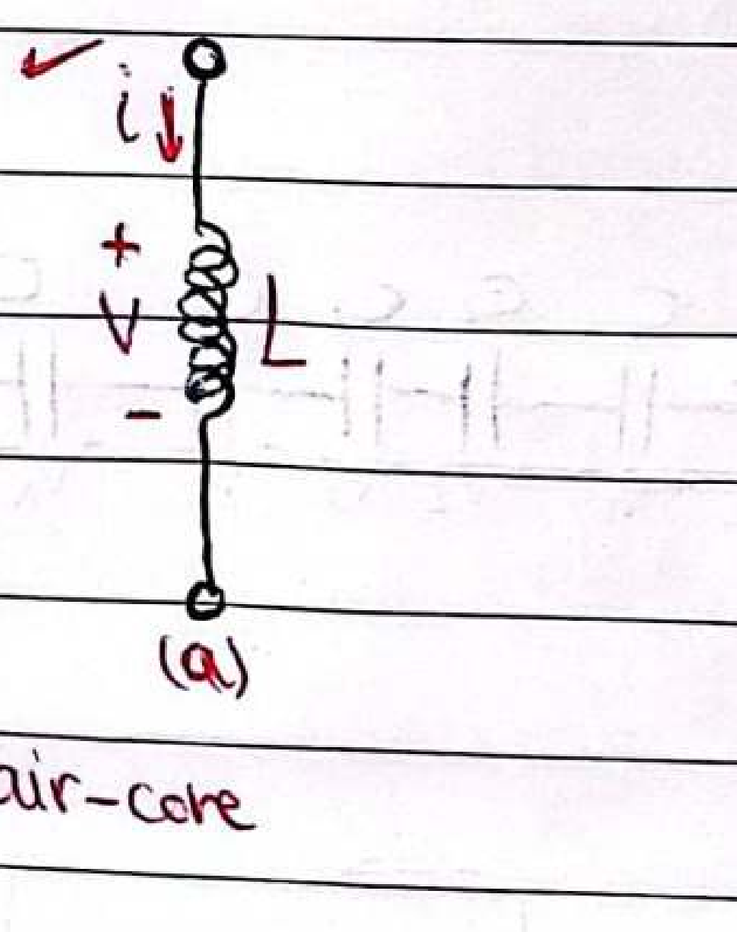
1) is a passive element designed to store energy in its magnetic field.

2) used in power supplies, transformers, radios, TVs, radars and electric motors.

Team ZerOne Computer & Networks Engineering

- An inductor : consists of a coil of conducting wire.

$$L = \frac{N^2 \mu A}{\ell}$$



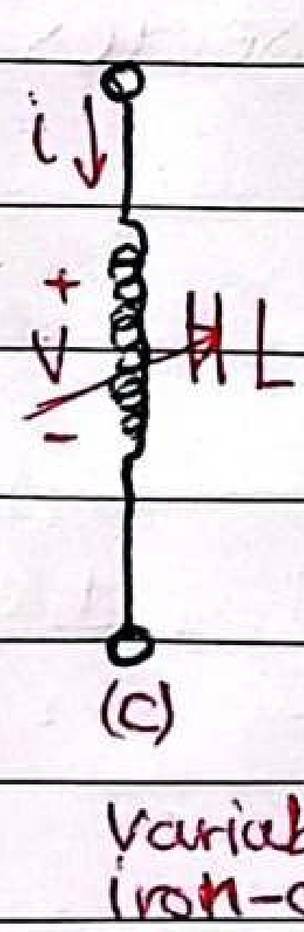
(a)

air-core



(b)

iron-core



(c)

variable iron-core

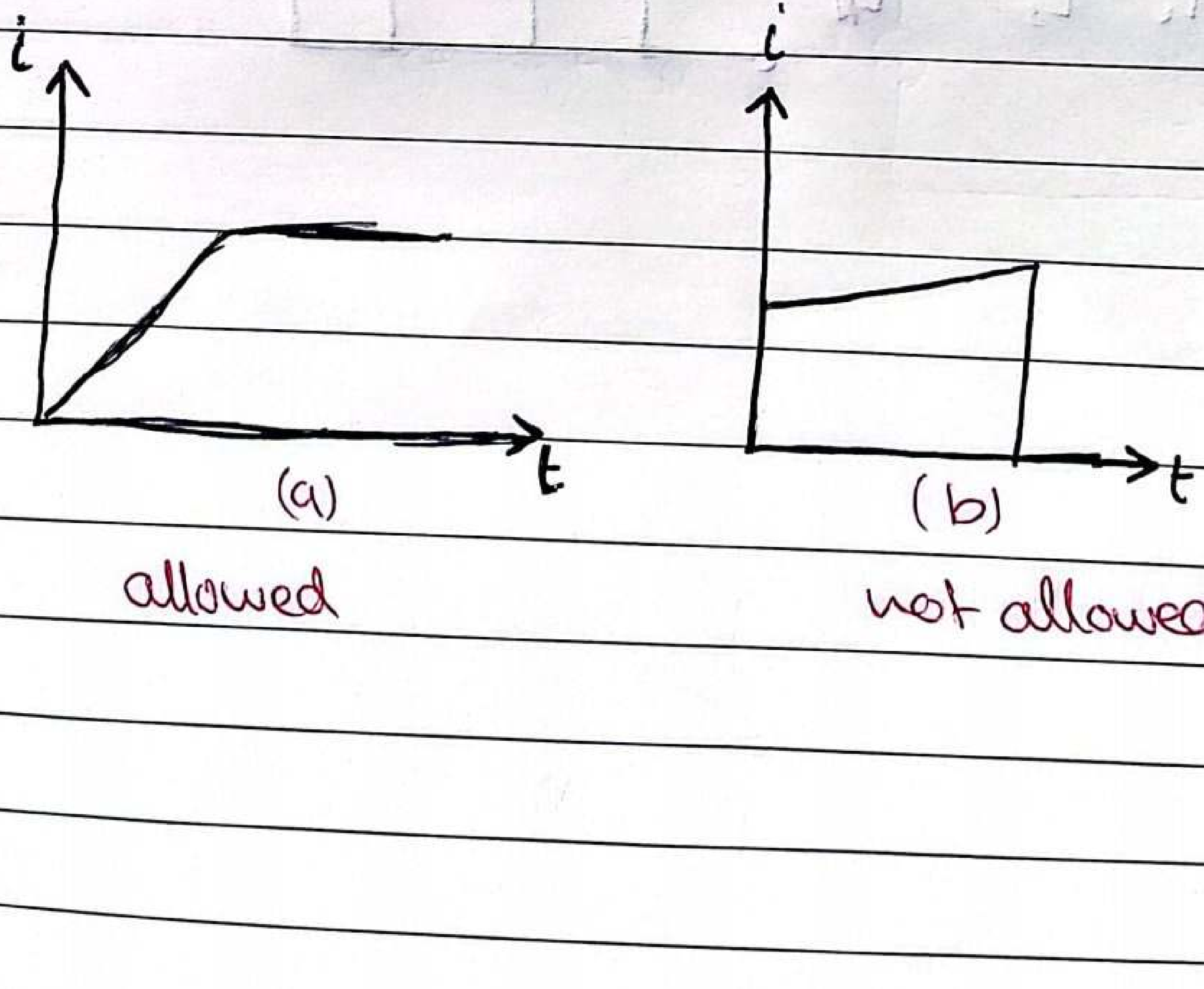
$$v = L \frac{di}{dt}$$

$$i = \frac{1}{L} \int_{t_0}^{t} v(t)\,dt + i(t_0)$$ ← التيار الحثي ذاخل الملف

$$W = \frac{1}{2} L i^2$$

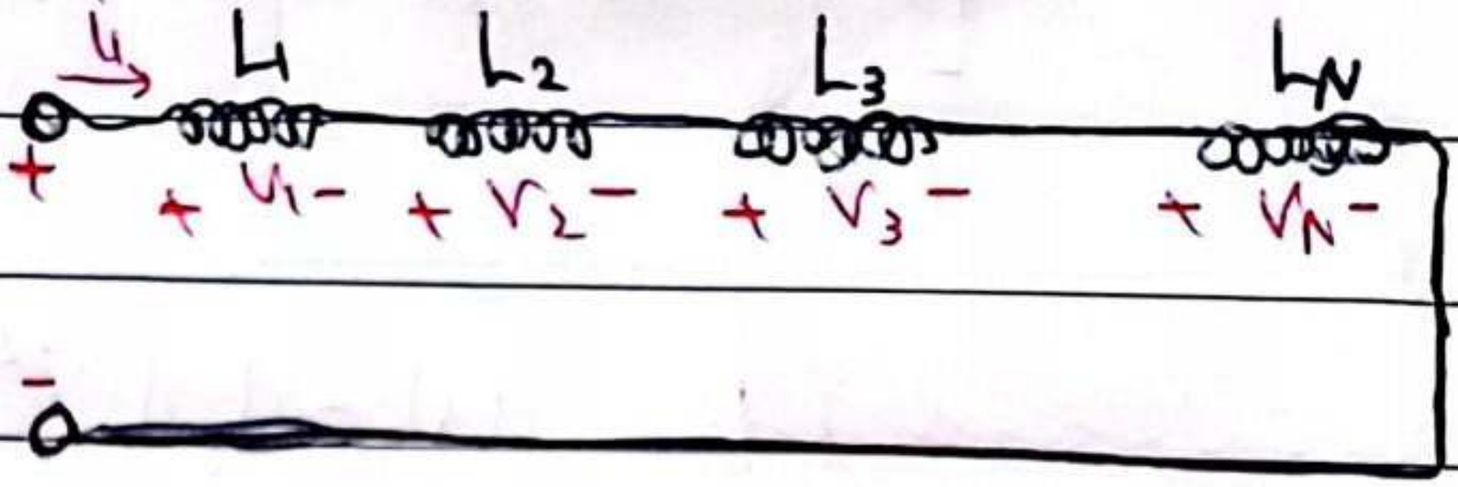
note :

- An inductor acts like a short circuit to dc
- The current through an inductor cannot change instantaneously.



series and parallel inductors:

series inductors:

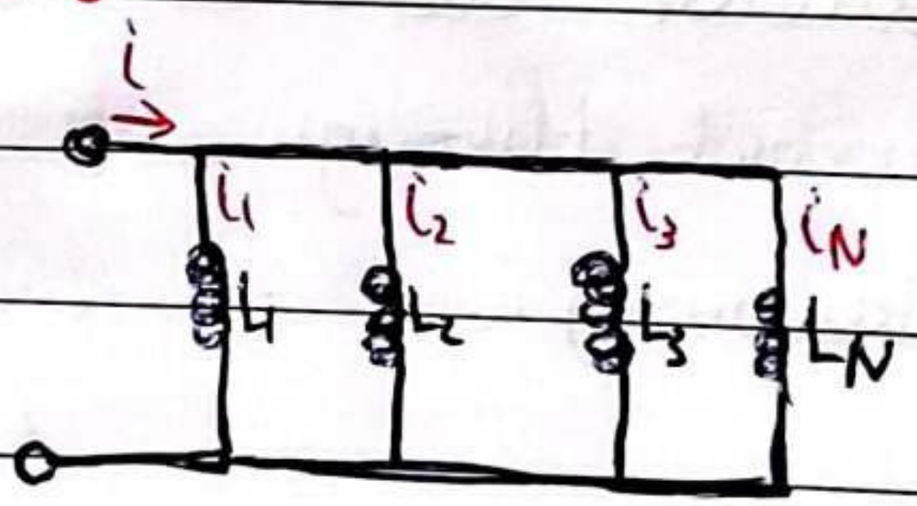


$$L_{eq} = L_1 + L_2 + L_3 + \cdots + L_N$$

Parcalled inductors:

Team
ZerOne
Computer & Networks Engineers

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \cdots + \frac{1}{L_N}$$



Ch.7 first-order circuits :

- A first order circuit is charactrized by a first order differential equation.

[The source free RC circuit]

- A source free RC circuit occurs when its dc source is suddenly disconnected. The energy already stored in the capaciter is released to the resistors.

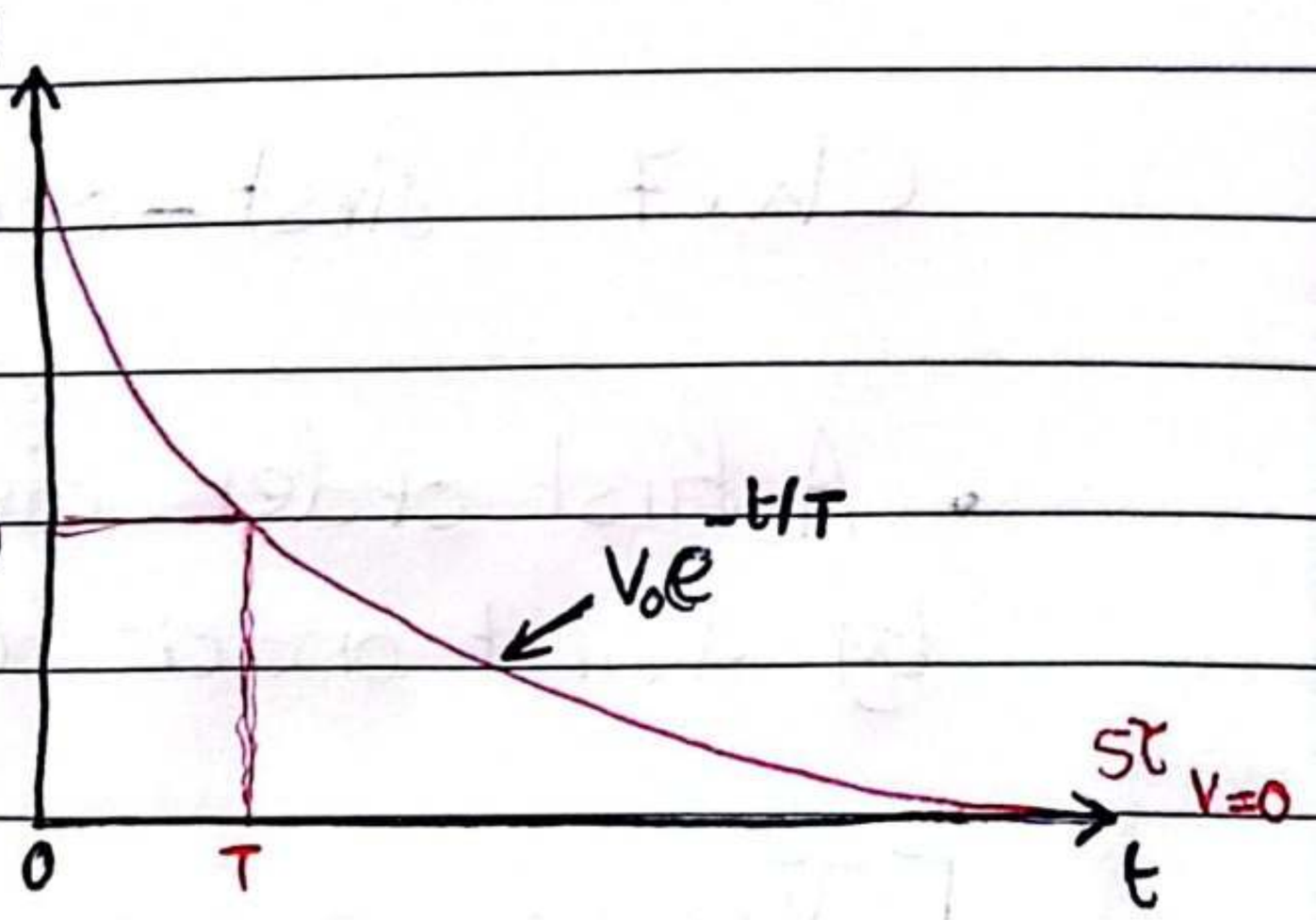
ZerOne
Computer & Networks Engineers

- Since the capacitor is initially charged by V_o

$$V(t) = V_o e^{-t/RC} \qquad \tau = RC$$

- The natural response: the behavior of the circuit itself with no external sources of excitation.

Time constant The time required to decay to a factor of 1/e or 36.8 percent of its initial value.



* The key to working with a source-Free RC Circuit is finding:

1) The initial voltage $V(0) = V_0$ across the capacitor
2) The time constant τ

ZerOne Computer & Networks Engineers

[The source-Free RL circuit]

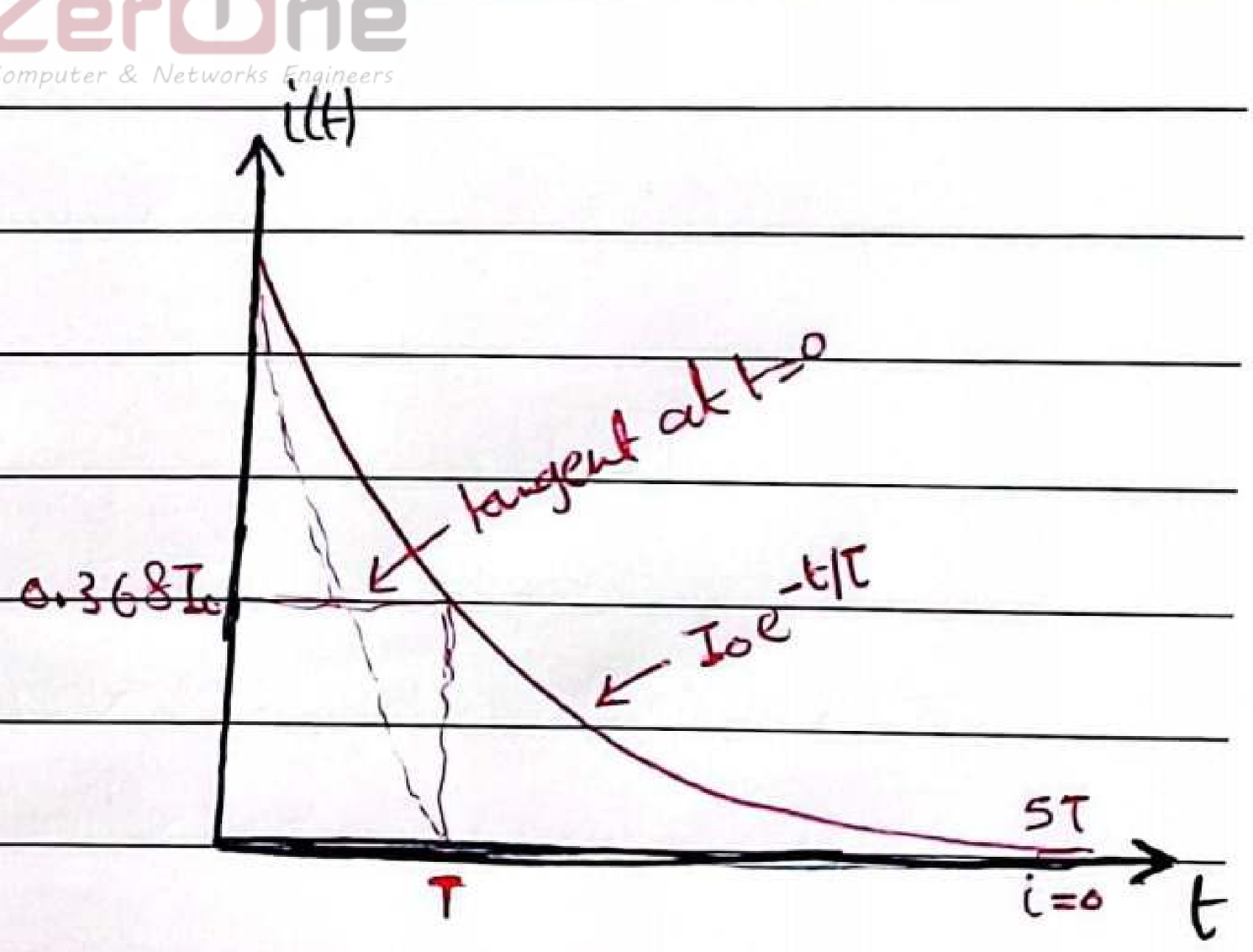
- determine the circuit response which we will assume to be the current i(t) through the inductor.

- assume that the inductor has an initial current I_0.

$$i(t) = I_0 e^{-Rt/L} \qquad \tau = \frac{L}{R}$$

$$i(t) = I_0 e^{-t/\tau}$$

Team ZerOne
Computer & Networks Engineers



- The key to working with a source-Free RL circuit is to find:

1) the initial current $i(0) = I_0$ through the inductor
2) the time constant τ of the circuit.

[step response of an RC circuit]

step response : is it's behavior when the excitation is the step function, which may be a voltage or a current source.

complete response = natural response + forced response
stored energy ZerOne Team independent source
Computer & Networks Engineers

transient response steady state response

- transient response : is the circuits temporary response that will die out with time.

- steady state response : is the behaviour of the circuit a long time after an external excitation is applied.

Source - Free

RC circuit

$\tau = RC$

$v(t) = V_0 e^{-\frac{t}{\tau}}$

steps:

$t<0$ عند:

$v(0)$ اوجد قيمة

capacitor → open circuit اجعل الـ -

$t>0$ عند:

اوجد Req

voltage → short اجعل -

current → open اجعل -

$\tau = Req\, C$ ثم اجد:

$v(t) = V_0 e^{-\frac{t}{\tau}}$ ثم اجد:

RL circuit

$\tau = \frac{L}{R}$

$i(t) = I_0 e^{-\frac{t}{\tau}}$

steps:

$t<0$ عند:

$i(0)$ اوجد قيمة

inductor → short circuit اجعل الـ -

$t>0$ عند:

اوجد Req

voltage → short اجعل -

current → open اجعل -

$\tau = \frac{L}{Req}$ ثم اجد:

$i(t) = i_0 e^{-\frac{t}{\tau}}$ ثم اجد:

Team ZerOne Computer & Networks Engineers

step response

RC circuit

$\tau = RC$

$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}}$

steps:

- عند $t<0$ اوجد قيمة $v(0)$
- capacitor → open اجعل ال -
- عند $t>0$ اوجد قيمة $v(\infty)$
- capacitor → open اجعل ال -
- اوجد R_{eq}
- Voltage → short اجعل ال -
- current → open اجعل ال -
- احسب $\tau = R_{eq}C$
- احسب $v(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}}$

RL circuit

$\tau = \frac{L}{R}$

$i(t) = i(\infty) + [i(0) - i(\infty)]e$

steps:

- عند $t<0$ اوجد قيمة $i(0)$
- inductor → short اجعل ال -
- عند $t>0$ اوجد قيمة $i(\infty)$
- inductor → short اجعل ال -
- اوجد R_{eq}
- Voltage → short اجعل ال -
- current → open اجعل ال -
- احسب $\tau = \frac{L}{R_{eq}}$
- احسب $i(t) = i(\infty) + [i(0) - i(\infty)]e^{-}$

Team ZerOne Computer & Networks Engineers

Note:

عند $t > 0$

مع مصدر

step response

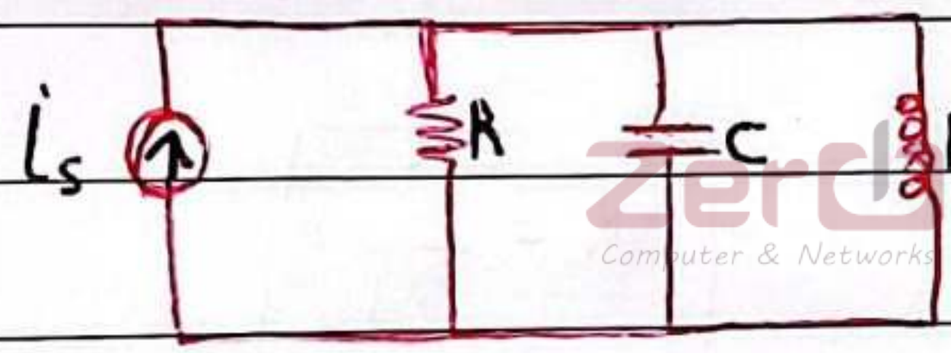
بدون مصدر

source free

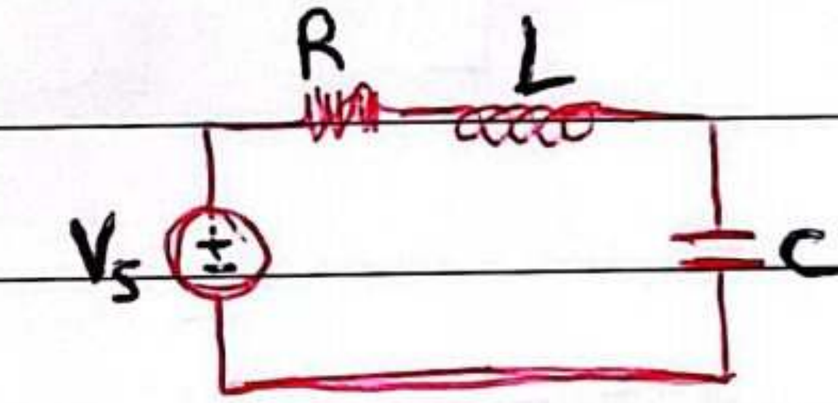
Team
ZerOne
Computer & Networks Engineers

ch.8 Second-order circuits :

A second-order circuit is characterized by a second-order differential equation. It consists of resistors and the equivalent of two energy storage elements.



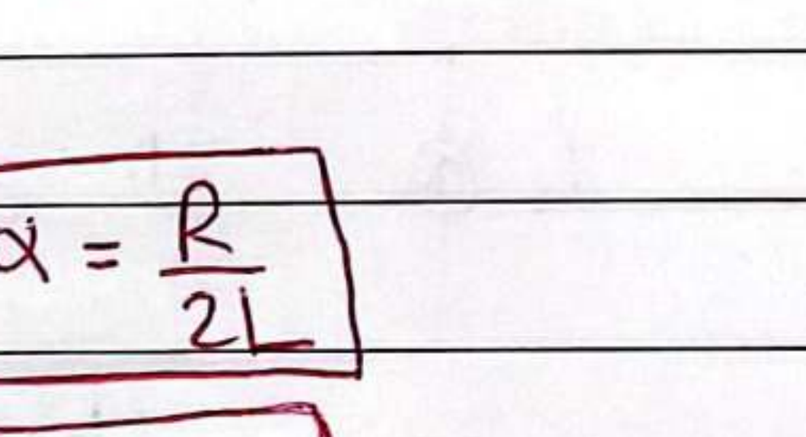
(b)



(a)

Team ZeroOne Computer & Networks Engineers

[The Source-Free Series RLC circuit]



$$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0$$

$$S^2 + \frac{R}{L}S + \frac{i}{LC} = 0 \quad \leftarrow \text{المعادلة المميزة}$$

$$S_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$$

$$S_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

$$\alpha = \frac{R}{2L}$$

$$\omega = \frac{1}{\sqrt{LC}}$$

Team ZerOne Computer & Networks Engineers

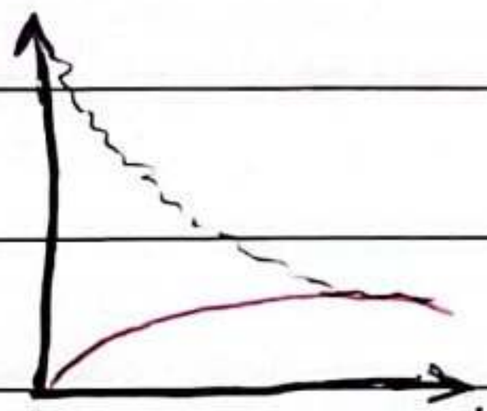
ω_0 → resonant frequency or undamped natural frequency

α → neper frequency or damping factor.

1. If $\alpha > \omega_0$, we have the overdamped case. $\Delta > 0$
2. If $\alpha = \omega_0$, we have the critically damped case. $\Delta = 0$
3. If $\alpha < \omega_0$, we have the underdamped case. $\Delta < 0$

1 Overdamped case ($\alpha > \omega_0$) :

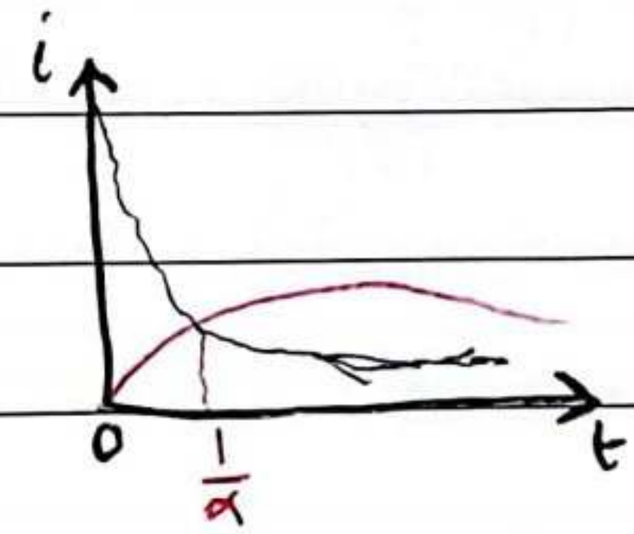
$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$



Team ZerOne Computer & Networks Engineers

2 Critically Damped Case ($\alpha = \omega_0$)

$$i(t) = (A_2 + A_1 t) e^{-\alpha t}$$

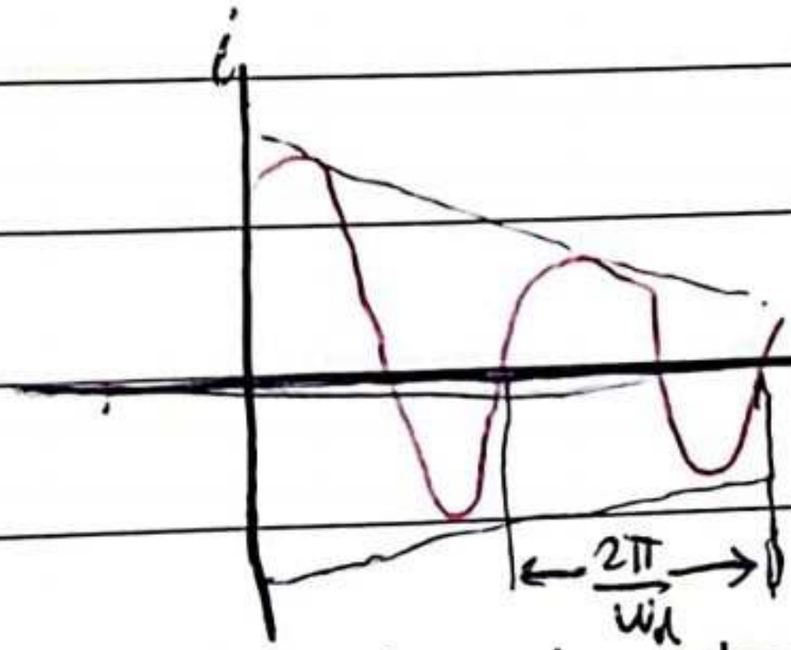


3 Underdamped Case ($\alpha < \omega_0$)

$$i(t) = (B_1 \cos \omega_d t + B_2 \sin \omega_d t) e^{-\alpha t}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

↳ damping frequency



note: طالبي الطلع B_2, B_1

* بطلع $i(0)$ عند $t<0$

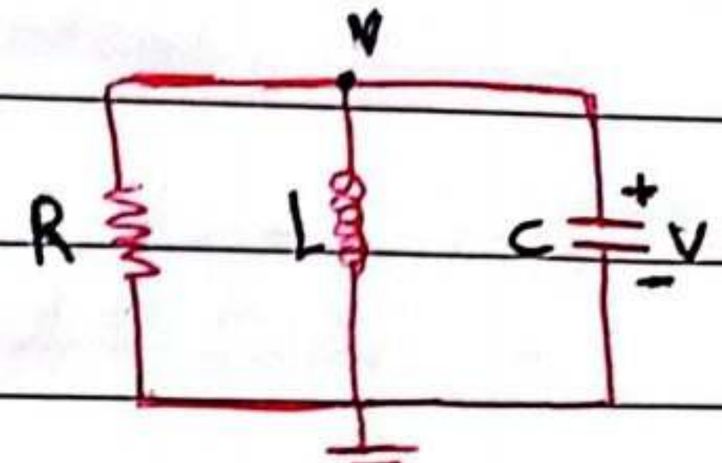
* بطلع $i'(0)$ عند $t=0$ وبتبدل كل إشارة V_0

$$i'(0) = -\frac{1}{L}(RI_0 + V_0)$$

Team ZerOne Computer & Networks Engineers

[Source Free Parallel RLC circuit]

$$\frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{1}{LC}v = 0$$



$$\equiv$$

$$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

$$\boxed{\alpha = \frac{1}{2RC}}$$

$$\boxed{\omega = \frac{1}{\sqrt{LC}}}$$

ZerOne Team
Computer & Networks Engineering

1. overdamped case ($\alpha > \omega_0$)

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

2. Critically Damped case ($\alpha = \omega_0$)

$$v(t) = (A + A_2 t) e^{-\alpha t}$$

3. Underdamped case ($\alpha < \omega_0$)

$$v(t) = (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

note: لإيجاد A_1 و A_2

* نوجد $V(0)$ و $V'(0)$

$V(0)$ لأن $t<0$

$V'(0)$ لأن $t>0$

قيمة التوازن

$$V(0) = -V_0 + RI_0$$

RC

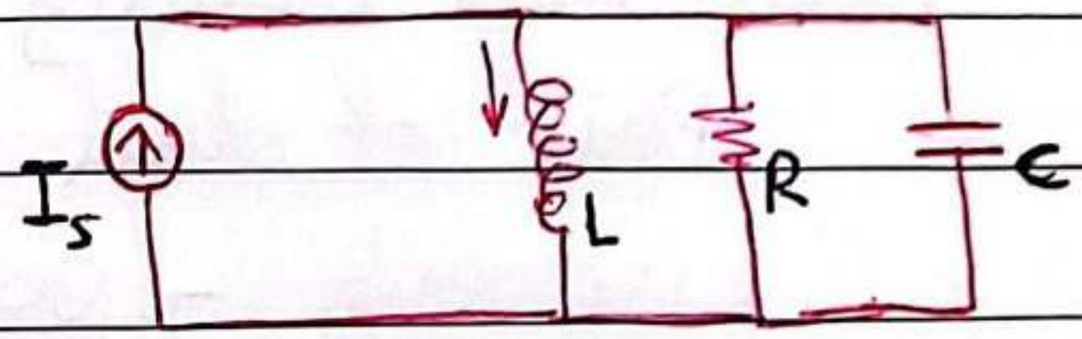
آخر شاحن بجسدك الكهرباء على طبيعي

Team
Zeroone
Computer & Networks Engineers

or

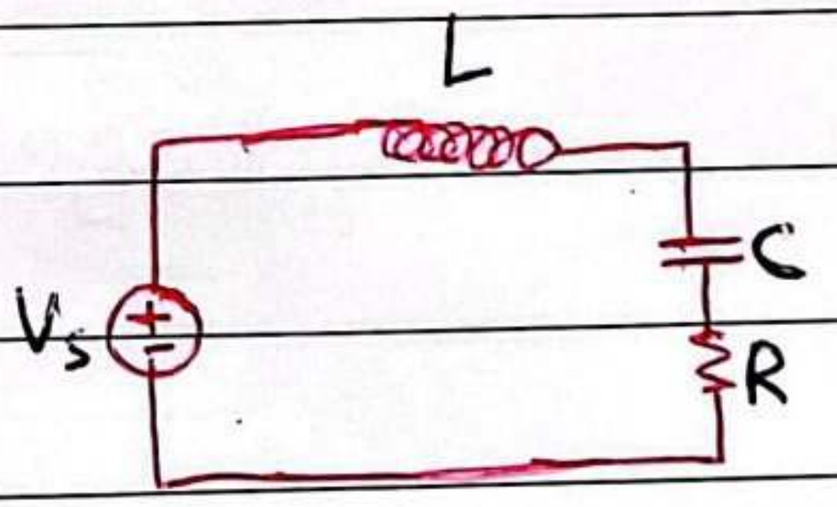
$$V'(0) = -\frac{1}{C}\left(\frac{V_0}{R} + I_0\right)$$

[RLC step response (parallel)]



$I_L(t) = I_s +$ equation.

[RLC step response (series)]



$V_c(t) = V_s +$ equation.

[Principle of Duality]

Duals and Duality :

Pair of dual terms.

current - voltage

open - short

L - C

R - G

series - Parallel

KCL - KVL

voltage source - current source.

Team ZeroOne Computer & Network Engineers